

Children's Diagnostic Center

Integrated Diagnostic Pathology Report

Accession:	CDC-2026-002910	Patient ID:	PID-44021
Report date:	2026-03-19	Age / Sex:	8 / M

Clinical history / indication

8-year-old male presenting with 3-week history of progressive morning headaches, ataxia, and one episode of emesis. Imaging revealed a 3.1 cm midline cerebellar vermis mass with obstructive hydrocephalus. Underwent posterior fossa craniotomy and subtotal resection. No history of Gorlin syndrome. No family history of childhood CNS neoplasm.

Specimen

Cerebellar vermis mass, subtotal resection (multiple fragments, aggregate 2.4 cm).

Macroscopic findings

Soft, friable, gray-pink tissue fragments. No grossly identifiable necrosis.

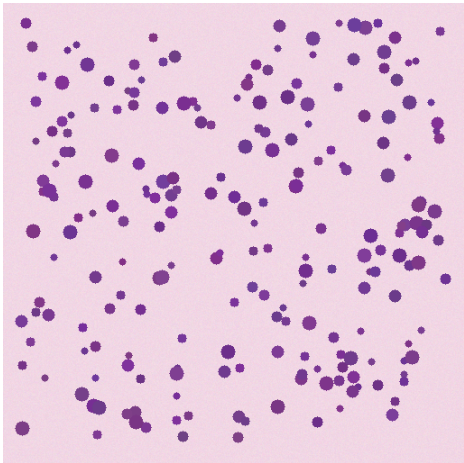
Microscopic / histopathology

Sections demonstrate a densely cellular small round blue cell tumor with classic medulloblastoma morphology. Tumor cells exhibit hyperchromatic nuclei, scant cytoplasm, and brisk mitotic activity. Homer Wright rosettes are identified focally. No large cell or anaplastic features are appreciated. No nodularity is seen. Surrounding cerebellar parenchyma shows reactive gliosis.

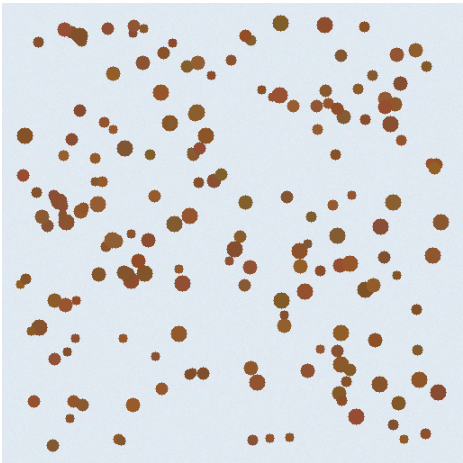
Immunohistochemistry

Marker	Result
Synaptophysin	Positive (diffuse)
NeuN	Focal positivity
INI1 (SMARCB1)	Retained nuclear expression
β-catenin	Cytoplasmic (no nuclear translocation)
GAB1	Positive (cytoplasmic, diffuse)
YAP1	Positive (nuclear and cytoplasmic)
p53	Wild-type pattern (weak, patchy)
Ki-67 (MIB-1)	Approximately 45% in hotspots

Representative images



H&E; ×20 — small round blue cell tumor



GAB1 IHC ×20

Molecular findings

Single-nucleotide variants / indels

Gene	HGVSc	HGVSp	VAF	Classification
PTCH1	c.2071C>T	p.R691*	0.49	Pathogenic
TP53	wild-type	wild-type		Wild-type

Structural variants / signatures

Kind	Description
rna_signature	SHH-pathway transcriptional signature (centroid match: SHH-activated)

Copy-number alterations

Gene	Copy number	Kind
MYC	2	diploid (no amplification)
MYCN	2	diploid (no amplification)

MSI status	MSS
TMB (mut/Mb)	1.1
Methylation class (DKFZ class)	MB (SHH (subgroup 3))

Pathologist comments

Morphology consistent with classic medulloblastoma. IHC profile (GAB1+, YAP1+, β -catenin cytoplasmic) supports SHH activation; methylation classifier confirms MB, SHH subgroup. TP53 wild-type and no MYC/MYCN amplification — favorable molecular risk relative to TP53-mutant SHH subtype. INI1 retained excludes AT/RT. Recommend craniospinal staging and integrated risk stratification per current pediatric MB protocols.

Electronically signed: **Reporting Pathologist, MD, PhD**
2026-03-19